

Jinhyeok Kim

Artificial Intelligence Graduate | Applied AI Developer | Data Analyst

Sydney, NSW, Australia | 0468 544 027 | winsuwon127@gmail.com | Portfolio: jinhyeok-portfolio-amber.vercel.app | GitHub: github.com/Jinhyeok513

Career Profile

AI graduate with hands-on academic and portfolio experience in NLP, computer vision, data preparation, model evaluation, Excel-based planning and project documentation. The CV presents a broader record for graduate roles across data, AI, analytics and technology-enabled operations.

Career Objective

Build a graduate career in applied AI and data analysis, using structured evidence, reliable reporting and practical software delivery to support business and technical teams.

Technical and Business Competencies

Python | Pandas | NumPy | SQL | Data cleaning

Data validation | Machine learning | NLP | Computer vision | Microsoft Excel - Intermediate

PivotTables | XLOOKUP/VLOOKUP | SUMIFS/COUNTIFS | INDEX-MATCH | Analytical reporting

Project documentation | Git/GitHub | Streamlit

Education

Bachelor of Artificial Intelligence - Completed 2026

University of Technology Sydney (UTS), Sydney, Australia

- Graduated with Distinction; WAM 77.94 and GPA 5.88/7.00.
- Relevant study included data analytics, machine learning, computer vision, generative AI, software development and business information systems.

Diploma of Information Technology - 2023

UTS College, Sydney, Australia

- Built foundations in Python and Java programming, web systems, system analysis, technical documentation and business information systems.

Relevant Coursework

Introduction to Data Analytics; Advanced Data Analytics Algorithms, Machine Learning; Design, Data, and Decisions; Project Management and the Professional; AI/Analytics Capstone Project; Communication for IT Professionals; Data Structures and Algorithms; Image Processing and Pattern Recognition; Deep Learning and Convolutional Neural Network; Introduction to Computational Intelligence; Introduction to Artificial Intelligence; The Ethics of Data and AI

Applied Academic Evidence

- Data analytics coursework covered cleaning, validation, statistical interpretation and communication of findings from structured datasets.
- Advanced analytics and machine-learning study developed model-comparison discipline, feature preparation, evaluation and limitation reporting.
- Design, Data, and Decisions strengthened Excel-based analysis using PivotTables, lookup functions, conditional formatting, charts and practical decision support.
- Project Management and the Professional developed WBS, AON, Gantt scheduling, risk, stakeholder, budget and human-resource planning capability.
- Communication for IT Professionals supported clear technical documentation, report structure and audience-aware presentation of analysis.

Detailed Project Experience

BioGeoDA - Australian Plant Data Integration and AI-Assisted Trait Extraction | 2025-2026

UTS AI/Analytics capstone team project and public portfolio reconstruction | Python | Pandas | NLP | TF-IDF | Logistic Regression | BERT QA | Streamlit | GitHub

Problem: botanical traits were scattered across structured datasets and OCR-ready journal text. Approach: combine cleaning, standardisation, NLP baselines and validation to make outputs reviewable.

- Cleaned and standardised multi-source plant datasets by reviewing missing values, duplicate records, inconsistent species names and mixed trait formats.
- Developed keyword-tagging and fuzzy-matching workflows to locate species in OCR-processed botanical text while retaining source, page, paragraph and context metadata.
- Worked across TF-IDF classification and BERT-based question-answering pipelines for keyword-dependent and context-dependent trait extraction.
- Contributed to validating 844 new trait records from more than 2,400 AI-generated candidates across a 1,674-species project checklist.

AI Gym Trainer - Exercise Video Analysis Application | 2025-2026

Portfolio AI application | Python | Next.js | FastAPI | MobileNetV2 | MediaPipe Pose | OpenCV | CSV/JSON export

Problem: exercise videos need interpretable analysis rather than a single opaque classification label. Approach: combine video preprocessing, model inference, pose landmarks and structured reporting.

- Built an end-to-end video analysis workflow covering upload, preprocessing, exercise classification, pose extraction, annotated video output and session reporting.
- Integrated MobileNetV2 classification with MediaPipe Pose landmarks, OpenCV rendering, joint-angle calculations and rule-based feedback.
- Documented model evidence honestly: MobileNetV2 reached 72.0% video accuracy and 78.18% video macro F1 on held-out comparison data, while a 10-video field test showed domain-shift limitations.

Tennis Ball Tracking and Trajectory Visualisation | 2025

Computer-vision portfolio project | Python | OpenCV | NumPy | Kalman filtering | FFmpeg | IoU/CLE evaluation

Problem: a tennis ball is small, fast and frequently obscured in broadcast footage. Approach: use classical detection signals plus Kalman smoothing and evaluation diagnostics.

- Implemented a Kalman-filter tracking pipeline using HSV colour cues, whiteness cues, optical flow, blobness scoring, court masking and net-region suppression.
- Packaged a 207-frame tracking output with representative frame captures, trajectory visualisation and frame-level diagnostics.
- Reported sparse-label evaluation limits transparently, including mean IoU 0.1827, Success@IoU ≥ 0.5 of 2.4% and mean CLE 131.18 px.

Event Management System Project Plan - Team Lead | 2025

Academic Project Management Case Study | Project Management | Excel | WBS | AON | Gantt scheduling | Risk and budget planning

Problem: a complex event-system scenario needed a credible plan rather than a built product. Approach: lead planning, schedule, scope, resource, risk and budget documentation for an academic simulation.

- Led a seven-member academic team in developing an end-to-end project plan for a proposed AUD 1.5 million Event Management System supporting a major Sydney festival scenario.
- Directed task allocation, reviewed team deliverables and integrated sections covering scope, stakeholders, risk, quality, budget, resources and change control.
- Coordinated an Excel-based schedule and WBS with more than 100 tasks, dependencies, milestones, durations, float and critical-path analysis across an 18-month plan.

Leadership and Collaboration

- Led a seven-member academic team by setting direction, assigning tasks, reviewing sections and integrating the final project-management report.
- Worked in a four-person UTS capstone team while separately publishing a portfolio-safe reconstruction of the AI/NLP contribution.
- Communicated technical project results through reports, README documentation, demos and portfolio case-study writing.

Role Fit and Transferable Strengths

- Can contribute through python, pandas and numpy while continuing to build professional workplace experience.
- Comfortable working from imperfect information, checking source evidence and separating verified facts from assumptions.
- Able to translate technical outputs into concise documentation for business, operational and non-technical stakeholders.

- Brings Korean native language capability, professional English communication and HSK Level 4 Chinese for cross-cultural team environments.
- Open to learning role-specific platforms quickly, with a project record that shows practical adaptation across datasets, codebases and reporting requirements.

Languages

- Korean - Native
- English - Professional communication skills
- Chinese - HSK Level 4

Work Rights and Availability

Current Australian work-rights details and full-time availability available on request.